

Advanced Econometrics – Lab 05 – Mock test

Exercise 1¹

Let's analyse variable coding health status of respondents (*hstatus*). The variable takes three values: 1 for a person with bad health, 2 for moderate health, and 3 for good health status. Covariates for the dependent variable are: *income*, *female* – a binary variable (1 for women), *black*, and *num* – family size.

- a) Why shouldn't we use a logit/probit model for this problem?
- b) Estimate an ordered logit for health status with covariates mentioned above.
- c) Determine and interpret pseudo- R^2 statistics.
- d) Are variables in the model jointly significant?
- e) The parameter next to *income* is equal to $1.1578 * 10^{(-4)}$. Are we allowed to conclude on the basis of the value that *income* is insignificant?
- f) Having parameters estimated determine signs of marginal effects for *female* and *famsize* for the alternative *good health*.
- g) Perform goodness-of-fit tests.
- h) Determine and interpret marginal effects for the alternative *moderate health*.
- i) Determine and interpret marginal effects for the alternative *good health*.
- j) Verify hypothesis that *income* affects respondents' health linearly against the alternative that *income* has nonlinear, cubic impact. Use likelihood ratio test.

```
> ologit = ologit.reg(health~income+female+num, data=rd)
```

```
> summary(ologit)
```

Ordered Logit Regression

Log-Likelihood: -5294.072

No. Iterations: 5

McFadden's R2: 0.02912101

AIC: 10598.14

	Estimate	Std. error	t value	Pr(> t)	
income	1.1578e-04	7.0506e-06	16.4217	< 2.2e-16	***
female	-1.4122e-01	5.3507e-02	-2.6393	0.008308	**
num	2.9345e-02	1.2666e-02	2.3168	0.020515	*

----- Threshold Parameters -----

	Estimate	Std. error	t value	Pr(> t)	
Threshold (1->2)	-2.580795	0.092750	-27.8254	< 2.2e-16	***
Threshold (2->3)	-0.528933	0.077969	-6.7839	1.17e-11	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

¹This exercise uses *randdata.dta* dataset from A. Colin Cameron and Pravin K. Trivedi, *Microeconometrics: Methods and Application*, Cambridge University Press, New York, May 2005.

```
> # Joint significance
> # Likelihood ratio test
> ologit.unrestricted = polr(as.factor(health)~income+female+num, data=rd)
> ologit.restricted = polr(as.factor(health)~1, data=rd)
> lrtest(ologit.unrestricted, ologit.restricted)
Likelihood ratio test
```

```
Model 1: as.factor(health) ~ income + female + num
```

```
Model 2: as.factor(health) ~ 1
```

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	5	-5294.1			
2	2	-5452.9	-3	317.59	< 0.00000000000000022 ***

```
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
> # goodness-of-fit tests
> logitgof(rd$health, fitted(ologit.unrestricted), g = 10, ord = TRUE)
```

```
Hosmer and Lemeshow test (ordinal model)
```

```
data: rd$health, fitted(ologit.unrestricted)
```

```
X-squared = 102.25, df = 17, p-value = 0.00000000000003397
```

```
> pulkrob.chisq(ologit.unrestricted, c("female"))
```

```
Pulkstenis-Robinson chi-squared test
```

```
data: formula: as.factor(health) ~ income + female + num
```

```
X-squared = 17.99, df = 4, p-value = 0.00124
```

```
> # for Lipsitz et al. test
> rd$health = as.factor(rd$health)
> ologit.unrestricted = polr(health~income+female+num, data=rd)
> lipsitz.test(ologit.unrestricted)
```

```
Lipsitz goodness of fit test for ordinal response models
```

```
data: formula: health ~ income + female + num
```

```
LR statistic = 42.803, df = 9, p-value = 0.000002343
```

```

> # marginal effects
> options(scipen=999)
> margins.oglmx(ologit, atmeans=TRUE)
Marginal Effects on Pr(Outcome==1)
      Marg. Eff      Std. error  t value      Pr(>|t|)
female  0.00404525081  0.00153378879   2.6374      0.008354 **
income -0.00000334106  0.00000025207 -13.2545 < 0.000000000000000022 ***
num     -0.00084680153  0.00036781053  -2.3023      0.021320 *
-----
Marginal Effects on Pr(Outcome==2)
      Marg. Eff      Std. error  t value      Pr(>|t|)
female  0.01780157241  0.00671462027   2.6512      0.008021 **
income -0.00001466399  0.00000088939 -16.4877 < 0.000000000000000022 ***
num     -0.00371662855  0.00160455352  -2.3163      0.020542 *
-----
Marginal Effects on Pr(Outcome==3)
      Marg. Eff      Std. error  t value      Pr(>|t|)
female -0.0218468232  0.0082306515  -2.6543      0.007947 **
income  0.0000180050  0.0000010556  17.0560 < 0.000000000000000022 ***
num     0.0045634301  0.0019691219   2.3175      0.020477 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

> # likelihood ratio test
> model_R = polr(as.factor(health)~income+female+num, data=rd)
> model_U = polr(as.factor(health)~income+income2+income3+female+num, data=rd)
> anova(model_U, model_R)
Likelihood ratio tests of ordinal regression models

Response: as.factor(health)

      Model Resid. df Resid. Dev   Test
1      income + female + num      9166   10588.14
2 income + income2 + income3 + female + num      9164   10556.68 1 vs 2
  Df LR stat.      Pr(Chi)
1
2      2 31.46328 0.0000001471752
>

```